Image recognition neural network

MNIST Handwritten Digit Recognition Using Neural Networks

- **Course Project: Artificial Intelligence / Machine Learning**
- **Objective:** Build and compare three neural network architectures for handwritten digit recognition.
- **Models evaluated:**
  - Shallow Multi-Layer Perceptron (MLP)
  - Convolutional Neural Network (CNN)
  - Transformer Encoder
- **Dataset:** MNIST handwritten digits
- **Framework:** PyTorch and torchvision
- **Execution environment:** Google Colab, CPU

Problem Statement and Objectives

- The goal is to build a neural network that classifies handwritten digits from 0 to 9.
- Each image is a 28 × 28 grayscale image. The model receives the image and predicts one of ten digit classes.
- **Project objectives:**
  - Load the MNIST training and test datasets.
  - Implement three different neural network architectures.
  - Train each model using the training dataset.
  - Evaluate the models on the separate official test dataset.
  - Compare accuracy on clean and augmented images.
  - Analyze the strengths and limitations of each architecture.

MNIST Dataset

- MNIST is a widely used dataset for handwritten digit recognition.
- **Dataset characteristics:**
  - Total images: 70,000
  - Training images: 60,000
  - Test images: 10,000
  - Image dimensions: 28 × 28 pixels
  - Color channels: 1 (grayscale)
  - Number of classes: 10 (digits 0–9)
- **Data source:** torchvision.datasets.MNIST, which downloads MNIST through torchvision's configured sources.
- **Important experimental rule:** The models were evaluated using the separate test dataset rather than training on the test images.

Data Loading and Preprocessing

The experiments used 60,000 training examples and 10,000 test examples.

The images were converted into PyTorch tensors using transforms.ToTensor().

This converts pixel values to a numerical representation suitable for neural network processing, generally scaling them to the range 0–1.

Data pipeline:

MNIST images → Tensor conversion → Model training → Test evaluation

The training data was used to learn model parameters. The official test data was used to measure generalization to unseen examples.

The optional augmented-image evaluation measured how well the trained models handled transformed images.

Model 1 — Shallow Multi-Layer Perceptron (MLP)

- The MLP is a fully connected neural network that treats each image as a vector.
- **Architecture:**
  - 28 × 28 image
  - Flatten into 784 pixel values
  - Fully connected layer: 256 neurons
  - ReLU activation
  - Dropout
  - Output layer: 10 classes
  - Predicted digit (0–9)
- **How it works:**
  - The image is flattened into a one-dimensional vector containing 784 pixel values. Fully connected layers learn patterns that help distinguish the ten digit classes.
- **Advantages:**
  - Simple architecture
  - Fast to train
  - Strong baseline for MNIST
- **Limitation:**
  - Flattening the image does not explicitly preserve the spatial relationships between neighboring pixels.

Model 2 — Convolutional Neural Network (CNN)

- A CNN is designed to learn spatial patterns in images.
- **Architecture:**
  - 28 × 28 grayscale image
  - Convolutional layer: 32 filters
  - ReLU activation
  - Max pooling
  - Convolutional layer: 64 filters
  - ReLU activation
  - Max pooling
  - Flatten
  - Fully connected layer: 128 neurons
  - Dropout
  - Output layer: 10 classes
- **How it works:**
    - Convolutional filters learn local features such as edges, curves, and digit strokes. Pooling reduces the spatial dimensions of the feature maps.
    - The final fully connected layers use these learned features to classify the digit.

Advantages:

- Preserves spatial structure
- Learns useful visual features
- Achieved the best accuracy in this experiment

Model 3 — Transformer Encoder

- The Transformer model uses self-attention to learn relationships between different regions of an image.
- **Architecture concept:**
  - 28 × 28 image
  - Divide image into patches
  - Convert patches into tokens
  - Add positional information and a class token
  - Transformer encoder
  - Classification layer
  - Predicted digit (0–9)

  How it works:

  The image is divided into small patches. Each patch is represented as a token, and self-attention helps the model learn relationships between different image regions.

  Advantages:

  - Learns relationships between image regions
  - Provides an opportunity to explore attention-based image recognition

  Limitations:

  - May require more training or tuning
  - Can be less efficient than a CNN for small image datasets

In this experiment, the Transformer was trained for three epochs.

Training Configuration

- All three models were trained using the CPU in Google Colab

Parameter	Configuration
Dataset	MNIST
Training examples	60,000
Test examples	10,000
Number of epochs	3
Batch size	128
Optimizer	Adam
Learning rate	0.001
Loss function	Cross-entropy
Evaluation	Official clean test set
Additional evaluation	Augmented-image test set

- **Training process:**
  - The model receives a batch of training images.
  - It predicts the digit class for each image.
  - Cross-entropy loss measures prediction error.
  - The optimizer updates the model's parameters.
  - The process repeats for each training batch and epoch.
  - After each epoch, the model is evaluated on the test set.

Experimental Results — Clean Test Accuracy

- The following results were obtained from the three trained models

Model	Epoch 1	Epoch 2	Epoch 3 / Final
MLP	96.22%	96.81%	97.55%
CNN	98.58%	98.48%	99.10%
Transformer	80.49%	92.66%	94.67%

- **Key observations:**
  - The CNN achieved the highest final clean-test accuracy of **99.10%**.
  - The MLP achieved **97.55%** accuracy, demonstrating that a relatively simple architecture can perform well on MNIST.
  - The Transformer achieved **94.67%** accuracy after three epochs.
  - The Transformer improved substantially from its first to its third epoch, suggesting that it was still learning useful representations.
- The results indicate that the CNN was the strongest model in this particular experiment.

Training Loss and Generalization

- Training loss decreased across the three epochs for all models

Model	Epoch 1 Training Loss	Epoch 3 Training Loss
MLP	0.5244	0.2018
CNN	0.3203	0.0889
Transformer	1.3898	0.4558

- **Interpretation:**
  - The MLP's training loss decreased steadily.
  - The CNN reached the lowest final training loss.
  - The Transformer also reduced its training loss substantially, but its final loss remained higher than those of the other models.
- The decreasing training loss indicates that the models learned patterns from the training data.
- The final test results show that the CNN generalized best to the official test set in this experiment.
- **Note:** Test loss did not decrease at every epoch for every model. This is not unusual; training and test performance can change differently during optimization.

Data Augmentation and Robustness

- Data augmentation introduces image transformations, such as small rotations and translations, to assess how robust a model is to changes in image appearance
- The experiment measured accuracy on an augmented version of the test images in addition to the official clean test set.

Model	Clean Test Accuracy	Augmented-Image Accuracy	Difference
MLP	97.55%	91.19%	−6.36 percentage points
CNN	99.10%	96.99%	−2.11 percentage points
Transformer	94.67%	82.98%	−11.69 percentage points

- **Key observations:**
  - The CNN achieved the highest accuracy on both clean and augmented images.
  - The MLP experienced a larger reduction in accuracy on augmented images.
  - The Transformer experienced the largest reduction in this experiment.
  - The CNN's smaller accuracy reduction suggests that it handled the tested transformations more robustly.
- **Important distinction:** Clean-test accuracy is the primary benchmark. Augmented-image accuracy is an additional robustness measurement and should not be substituted for the official test score.

Model Comparison

Criterion	MLP	CNN	Transformer
Final clean accuracy	97.55%	99.10%	94.67%
Augmented accuracy	91.19%	96.99%	82.98%
Learns spatial features	Indirectly	Yes	Through patch attention
Main operation	Fully connected layers	Convolution	Self-attention
Main strength	Simplicity	Image feature extraction	Relationships between patches
Main limitation	Loses explicit spatial structure when flattened	More complex than MLP	May need more training and tuning

Overall result: The CNN outperformed the MLP and Transformer in both clean-test accuracy and augmented-image accuracy under the tested configuration.

- The experiment highlights the importance of choosing a suitable architecture for image recognition.
- **Why did the CNN perform best?**
  - Convolutional layers learn local visual features and preserve the spatial structure of images. These properties are well suited to handwritten digit recognition.
- **Why did the MLP still perform well?**
  - MNIST is a relatively simple image classification problem. Even a shallow fully connected network can learn useful relationships between pixel values and digit classes.
- **Why might the Transformer have performed less well?**
  - The compact Transformer was trained for only three epochs. It may benefit from additional training, tuning, or a different patch configuration. Its lower accuracy does not establish that Transformers are generally inferior to CNNs.
- **What did augmentation reveal?**
  - All three models performed worse on transformed images than on clean images. The CNN experienced the smallest reduction, suggesting greater robustness to the transformations used.

Limitations and Future Improvements

- **Limitations:**
  - Each model was trained for only three epochs.
  - Training was performed on the CPU.
  - The architectures and hyperparameters were not exhaustively tuned.
  - Results are from the reported runs; multiple random-seed runs were not provided.
  - Augmented-image accuracy depends on the transformations applied.
- **Future improvements:**
  - Train the models for more epochs.
  - Compare training time and computational cost.
  - Tune the learning rate and model dimensions.
  - Test additional rotations and translations.
  - Plot training and test loss across epochs.
  - Display correctly classified and misclassified digits.
  - Repeat experiments with multiple random seeds.
- These experiments could help determine whether the observed differences persist under more extensive training and tuning.

Conclusion

- This project compared three neural network architectures for MNIST handwritten digit recognition: a shallow MLP, a CNN, and a Transformer encoder.
- The results were:
  - **CNN: 99.10% clean-test accuracy**
  - **MLP: 97.55% clean-test accuracy**
  - **Transformer: 94.67% clean-test accuracy**
- The CNN also achieved the highest augmented-image accuracy at 96.99%.
- The experiment demonstrates that convolutional neural networks are particularly effective for this handwritten digit classification task under the tested settings. It also shows how augmented-image evaluation can provide additional insight into model robustness.
- Future experiments with more training, tuning, and repeated runs could provide a more comprehensive comparison